
+24V_IN
VIN_FILT
VIN_FUSED
Q1_GATE
Q1 IRF9540N
D1 1.5KE36CA
R1 10k
R2 47k
C1 100nF
C2 470uF
C28 330uF/35V
L4 10uH
PWR_FLAG
+24V
GND
U1 TPS54560BDDA
VIN
EN
COMP
RT/CLK
BOOT
SW
FB
GND
BOOT1
C3 100nF
SW1
L2 33uH
+12V
COMP1
COMP1_R
C8 4.7nF
R4 10k
RT1
R3 100k
C9 47pF
D2 SS34
C4 10uF
C5 10uF
R5 6.49k
R6 21k
FB1
C6 100uF
C7 10uF
C11 10uF
U2 AP3211K
IN
SW
BS
FB
EN
C12 10uF
BOOT2
C10 100nF
SW2
L3 10uH
+5V
FB2
R9 100k
R10 2.32k
D3 SS34
C13 100uF
C14 10uF
U5 74HCT541
VCC
GND
VLV_IN1
VLV_IN2
VLV_IN3
VLV_IN4
VLV_IN5
VLV_IN6
VLV_EN
VLV_DRV1
VLV_DRV2
VLV_DRV3
VLV_DRV4
VLV_DRV5
VLV_DRV6
U3 BTS724G
U4 BTS724G
VBB
OUT1
OUT2
OUT3
OUT4
VALVE_BUB1
VALVE_BUB2
VALVE_BUB3
VALVE_BUB4
VALVE_INFL
VALVE_DEFL
D5 SS34
D6 SS34
D7 SS34
D8 SS34
D9 SS34
D10 SS34
R13 10k
R14 10k
R15 10k
R16 10k
J2 Conn_01x07_Pin
D11 LED
D12 LED
D13 LED
D14 LED
D15 LED
D16 LED
R23 4.7k
R24 4.7k
R25 4.7k
R26 4.7k
R27 4.7k
R28 4.7k
J7 Conn_01x04_Pin
+8V_SENS
SPEED_SIG
NEXT_8V
R39 100k
SP_DIV
R40 20k
J8 Conn_01x02_Pin
PRES_IN
D17 BZX84C3V3
R29 100k
R30 100k
BUT1
BUT2
R31 100k
R32 100k
BUT3
BUT4
R33 100k
R34 100k
ST_U3_12
ST_U3_34
R35 100k
R36 100k
ST_U4_12
ST_U4_34
R37 100k
R38 100k
INT_MCU
VLV_EN
R41 100k
PRES_SW
D18 BZX84C3V3
JP1 Jumper
EXT_8V
C30 10uF
C31 100nF
C32 100nF
C33 100nF
U11 78L08
U12 74LVC1G14
C35 100nF
D20 SMAJ12A
SPEED_SIG
U7 ESP32-DevKitC_30P
EN
IO36
IO39
IO34
IO35
IO32
IO33
IO25
IO26
IO27
IO14
IO12
IO13
GND
VIN
IO23
IO22
IO1
IO3
IO21
IO19
IO18
IO5
IO17
IO16
IO4
IO2
IO15
3V3
SPEED_PULSE
VLV_EN
INT_MCU
PRES_SW
DISP_MOSI
I2C_SCL
I2C_SDA
DISP_RES
DISP_SCLK
DISP_CS
DISP_DC
DISP_BL
C17 100nF
C18 10uF
R18 10k
R19 10k
DISP_CS
BUT5
C19 10uF
C20 100nF
C21 10uF
C22 100nF
C23 100nF
C24 100nF
C25 100nF
C26 100nF
C27 100nF
BUT1
BUT2
BUT3
BUT4
BUT5
J6 Conn_01x04_Pin
J4 Conn_01x04_Pin
J1 Conn_01x02_Pin
J3 Conn_01x06_Pin
J5 Conn_01x08_Pin
R20 4.7k
R21 4.7k
R22 1k
I2C_SDA
I2C_SCL
LED_A
D4 LED
U6 PCAL9555APW
SCL
SDA
INT
VDD
VSS
A0
A1
A2
IO0_0
IO0_1
IO0_2
IO0_3
IO0_4
IO0_5
IO0_6
IO0_7
IO1_0
IO1_1
IO1_2
IO1_3
IO1_4
IO1_5
IO1_6
IO1_7
ST_U3_12
ST_U3_34
ST_U4_12
ST_U4_34
BUT1
BUT2
BUT3
BUT4
BUT5
C29 100nF
Sheet: /
File: kamaz_leveler_hw.kicad_sch
Title:
Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.7
Id: 1/1